

Instructions

There are **50 total points**. This exam is closed-book and closed-notes except for **one 8.5 × 11-inch sheet of paper with handwritten notes (both sides permitted)**. No electronic devices, additional pages, or other materials are permitted. When asked to provide your answer within a figure or table, be careful to not exceed box boundaries. Bubbles must be filled out completely: ● is correct, ☑ ● ✗ are incorrect. All answers must be given within the provided circles, answer boxes, figures or tables. Write your full name in the box to acknowledge the instructions.

Linear Regression

1. [2 points]: In the linear regression hands-on, we modeled packets vs. bytes per flow but had high error. Why did the linear fit perform poorly?

- | | |
|---|---|
| ☐ HTTP traffic is encrypted | ☐ Flows mix large data and small ACK packets |
| ☐ The pcap file was corrupted | ☐ Linear regression can't handle traffic data |

2. [2 points]: Suggest one approach to improve the accuracy of predicting bytes from packets in the HTTP traffic regression task.

(Answer inside the box)

3. [3 points]: When would you use logistic regression instead of linear regression for network traffic analysis?

- | | |
|---|---|
| ☐ When input features are categorical | ☐ When output is binary (attack vs. normal) |
| ☐ When the dataset is very large | ☐ When you need faster training time |

Decision Trees and Ensemble Methods

4. [2 points]: Which are drawbacks of decision trees that random forests address? (Select all that apply.)

- ☐ Prone to overfitting training data
- ☐ Cannot handle categorical features
- ☐ High variance (small data changes → different trees)
- ☐ Too slow to train on large datasets

Initials: _____

5. [3 points]: Describe **two specific ways** that random forests introduce randomness to create diverse trees in the ensemble.

(Answer inside the box)

6. [3 points]: In the IoT privacy hands-on with the Nest camera, what traffic feature made classifying motion events easy?

☐ Packet header checksums

☐ Traffic volume changes correlated with events

☐ DNS query patterns

☐ TCP sequence numbers

Neural Networks and Deep Learning

7. [2 points]: Deep neural networks always achieve higher accuracy than traditional ML models (like random forests) on network traffic classification tasks.

☐ Yes ☐ No

8. [2 points]: Give an example of a spurious correlation that a deep learning model might learn from network traffic data, leading to poor generalization.

(Answer inside the box)

9. [3 points]: Why might a traditional ML model (like random forests) with engineered features outperform a deep neural network for DDoS detection? (Select all that apply.)

☐ Engineered features capture distinct attack patterns ☐ Deep learning can only process raw network traffic

☐ Limited training data causes deep learning to overfit ☐ Random forests are always more accurate

10. [3 points]: What is an advantage of using deep learning with nPrint over traditional ML with engineered features? (Select all that apply.)

☐ Same representation works across multiple tasks ☐ Always achieves higher accuracy

☐ No manual feature engineering needed per task ☐ Requires less training data

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11. [2 points]: What is a potential benefit of using nPrint's standardized bitmap representation for network traffic compared to manual feature engineering?

- ☐ It always produces better classification accuracy ☐ It ensures reproducibility across different researchers
☐ It requires less computational resources ☐ It eliminates the need for deep learning

12. [3 points]: Explain why nPrint uses a three-valued representation (1, 0, -1) for packet bits instead of just binary (1, 0).

(Answer inside the box)

Unsupervised Learning and Anomaly Detection

13. [3 points]: In the anomaly detection hands-on activity using the Log4j dataset, attack traffic represents a small fraction of total traffic (rare events). Explain why K-means clustering struggles to detect such rare attack traffic as anomalies.

(Answer inside the box)

14. [2 points]: DBSCAN can identify anomalies as points that do not belong to any cluster, while K-means will always assign every point to a cluster.

- ☐ Yes ☐ No

15. [2 points]: Why can unsupervised learning detect previously unseen attacks that supervised classifiers would typically miss? (Select all that apply.)

- ☐ Unsupervised models are more accurate ☐ Novel attacks can't exist in labeled training data
☐ Unsupervised methods detect deviations from normal ☒ Supervised models only work on encrypted traffic

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Diffusion Models and Synthetic Traffic Generation

16. [3 points]: What is a limitation of NetDiffusion for generating synthetic traffic? (Select all that apply.)

- ☐ Does not capture timing or temporal relationships ☐ Cannot generate valid packet headers
☐ Needs post-processing for checksums/seq numbers ☐ Only works for encrypted traffic

17. [3 points]: In synthetic traffic generation, *fidelity* measures how closely synthetic data matches real traffic, while *diversity* measures how much variation the synthetic data provides. Explain why both extremes of this trade-off are problematic.

(Answer inside the box)

18. [2 points]: In NetDiffusion's post-processing step, which dependency tree rules must be applied to ensure generated traffic is protocol-compliant? (Select all that apply.)

- ☐ Inter-packet rules like TCP sequence numbers ☐ Encryption of all packet payloads
☐ Random assignment of source IPs ☐ Intra-packet rules like checksum consistency

19. [3 points]: Why is synthetic traffic useful for ML-based classifiers? (Select all that apply.)

- ☐ Synthetic is always more accurate than real ☐ Real traffic contains private/sensitive info
☐ Can generate rare attacks hard to capture ☐ Eliminates need for any real training data

Feedback

20. [1 point]: Interest (1=Boring!; 10=Amazing!):

Difficulty (1=Too easy; 10=Too hard):

21. [1 point]: 1. One topic you found most interesting. 2. One suggestion for improvement:

(Answer inside the box)

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